

Total No. of Questions : 8]

SEAT No. :

P827

[5870]-1150

[Total No. of Pages : 2

T.E. (I.T.)

**WEB APPLICATION DEVELOPMENT
(2019 Pattern) (Semester - II) (414464A)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What is MVC? Explain MVC architecture in detail. [6]
b) Explain event binding and property binding in angular with example. [6]
c) Explain different types of Hooks in ReactJS. [6]

OR

- Q2)** a) What is TypeScript? List advantages & disadvantages of using it. [6]
b) What is Pipe? Explain with example. [6]
c) Explain Redux - Architecture in detail. [6]

- Q3)** a) Explain ExpressJS as middleware with example. [5]
b) What is NoSQL? Explain different features of MongoDB. [6]
c) Explain callbacks in NodeJS with suitable example. [6]

OR

- Q4)** a) Explain concept of routes in express with example. [5]
b) Write code to create collection, insert data & delete data in MongoDB using NodeJS. [6]
c) Explain NodeJS events with example. [6]

P.T.O.

- Q5)** a) What is Mobile-First and Mobile Web? List different Mobile Devices. [6]
b) What is jQuery Mobile? Explain Mobile jQuery Framework in detail. [6]
c) Explain any 3 events in jQuery Mobile with example? [6]

OR

- Q6)** a) Compare jQuery with jQuery Mobile. [6]
b) What is jQuery Mobile? What are the advantages and disadvantages of jQuery Mobile? [6]
c) Write a code to create a header and footer in jQuery Mobile. [6]
- Q7)** a) Explain in detail types of Elastic Load Balancer. [6]
b) What are the different components of VPC? [5]
c) What is elastic beanstalk and enlist the advantages of using it? [6]

OR

- Q8)** a) Explain any three AWS Storage services. [6]
b) What is Elastic Load Balancer and explain its working. [5]
c) What is AWS cloud? List different services provided by it. [6]

Total No. of Questions : 8]

SEAT No. :

PA-1509

[Total No. of Pages : 2

[5926]-129

T.E. (Information Technology)
WEB APPLICATION DEVELOPMENT
(2019 Pattern) (Semester - II) (414464A)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer four questions.
- 2) Figures to the right indicate full marks.
- 3) Assume suitable data if necessary.
- 4) Neat diagrams must be drawn wherever necessary.

- Q1)** a) List and explain different types of structural directives in Angular. [6]
b) How would you demonstrate the term web framework? Give the reason for using a web framework. [6]
c) What is Angular JS? Explain its features. [6]

OR

- Q2)** a) List and explain the features of any three popular web frameworks. [6]
b) Explain MVC architecture with a suitable diagram. [6]
c) How would you use the term typescript? Give the advantages and disadvantages of using it. [6]

- Q3)** a) List and explain the features of advanced MongoDB. [6]
b) Explain any four methods of console object in node.js with suitable examples. [6]
c) Write a short note on PM2 microservices. [5]

OR

- Q4)** a) Explain the callbacks in node.js with a suitable example. [6]
b) What is the purpose of map reduce? Explain it with a suitable example. [6]
c) Write a short note on Mongoose ODM [5]

P.T.O.

- Q5)** a) Explain the mobile devices and desktop devices in detail. [6]
b) What is j Query? Explain its architecture in detail. [6]
c) What is CDN? How it is powerful in web development? [6]

OR

- Q6)** a) Explain the mobile devices and desktop devices in detail. [6]
b) What is j Query? Is j Query a mobile framework? If yes, explain in brief [6]
c) What is a page? Write a code to create a page in j Query mobile. [6]

- Q7)** a) What is VPC? Explain the components of VPC. [6]
b) What is cloud computing? What are the benefits of cloud computing? [6]
c) List and explain the steps to deploy the application on the elastic beanstalk. [5]

OR

- Q8)** a) What are the storage services provided by AWS? [6]
b) What is S3 bucket and how to create a bucket? [6]
c) What is PuTTY? How to connect the EC2 instance with PuTTY? [5]



Total No. of Questions : 8]

SEAT No. :

P491

[6003]-712

[Total No. of Pages : 2

T.E.I.T.

**WEB APPLICATION DEVELOPMENT
(2019 Pattern) (Semester -II) (414464A)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer four questions.
- 2) Figures to right indicate full marks.
- 3) Assume suitable data, if necessary.
- 4) Neat diagram must be drawn wherever necessary.

- Q1)** a) Explain MVC architecture with a suitable diagram. [6]
b) List and explain different types of structural directives in Angular. [6]
c) What way would you design simple application in typescript to demonstrate the use of modules? [6]

OR

- Q2)** a) List and explain the features of any three popular web frameworks. [6]
b) How would you use the term typescript? Give the advantages and disadvantages of using it? [6]
c) Explain any 2 Hooks in React JS with example. [6]

- Q3)** a) Write and explain a simple application using REST HTTP Method API in node JS. [6]
b) Explain how to perform CRUD operations in a Node JS application. Provide examples of CRUD implementation. [6]
c) Write a short note on PM2 microservices. [5]

OR

- Q4)** a) What is template engine? How to create and use it using Express.JS?[6]
b) List and explain the features of advanced MongoDB. [6]
c) Explain the role of NPM (Node Package Manager) in Node.js development. [5]

P.T.O.

- Q5)** a) What is jQuery Mobile? Explain Mobile jQuery framework in detail. [6]
b) Explain the mobile devices and desktop devices in detail ? [6]
c) Write a code to create a header and footer in jQuery Mobile. [6]

OR

- Q6)** a) Explain any 3 events in jQuery Mobile with example? [6]
b) Compare jQuery with jQuery Mobile. [6]
c) Explain the concept of Mobile-First design in web development. Explain its significance and benefits in creating responsive and user-friendly websites. [6]
- Q7)** a) Explain the features and benefits of AWS Cloud. [6]
b) What is S3 bucket and how to create a bucket? [6]
c) Explain the different components of VPC? [5]

OR

- Q8)** a) What is PuTTY? How to connect EC2 instance with PuTTY? [6]
b) Explain the process of deploying a website or web application on AWS. [6]
c) What is Elastic Load Balancer and explain its working. [5]

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Total No. of Questions : 8]

SEAT No. :

P-7627

[Total No. of Pages : 2

[6180]-147

T.E. (IT)

**WEB APPLICATION DEVELOPMENT
(2019 Pattern) (Semester - II) (314464A)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6 and Q7 and Q8.
- 2) Figures to the right side indicate full marks.
- 3) Use of nonprogrammable calculator is allowed.
- 4) Assume suitable data if necessary.

Q1) a) Write a simple application in typescript to demonstrate the use of modules. **[9]**

b) What is pipe? Demonstrate the code for pipes in Angular. **[9]**

OR

Q2) a) Give the simple layout of the Angular application with multiple components. Explain how to create and use components in Angular? **[9]**

b) Explain the basic hooks in React JS. Explain any two hooks in brief. **[9]**

Q3) a) What is CRUDE? Explain the crude CRUDE in node.JS. **[9]**

b) What is node.JS? Explain file handling in node.js. **[8]**

OR

Q4) a) Write a code using Express.JS to illustrate the concept of roots. **[9]**

b) What is replication? Enlist the advantages of replication in database system. **[8]**

Q5) a) What is navigation? Write a code to navigate from one page to another page in jQuery mobile. **[9]**

b) What is jQuery mobile? Explain layout in jQuery Mobile design with its types. **[9]**

P.T.O.

OR

- Q6)** a) Why CSS is important in designing mobile websites? Give an example of CSS class used for mobile website design. [9]
b) List any five widgets in jQuery mobile and explain any two in brief. [9]

- Q7)** a) What is EC2 service? Explain Steps to deploy website on EC2. [9]
b) What is AWS cloud? What are the services provided by AWS? Explain any two in brief. [8]

OR

- Q8)** a) What is ELB? What are the ELB types? List advantages of ELB. [9]
b) What is VPC? What are the components of VPC? [8]



Total No. of Questions : 8]

SEAT No. :

PB-3867

[Total No. of Pages : 2

[6262]-131

T.E. (Information Technology)
Web Application Development
(2019 Pattern) (Semester - II) (314453 (A))

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6 and Q7 or Q8
- 2) Figures to the right side indicate full marks.

Q1) a) Write a simple application in typescript to demonstrate the use of modules. [9]

b) What is pipe? Demonstrate the code for pipes in Angular. [9]

OR

Q2) a) Give the simple layout of the Angular application with multiple components. Explain how to create and use components in Angular?[9]

b) Explain the basic hooks in React JS. Explain any two hooks in brief.[9]

Q3) a) What is CRUD? Explain the CRUD using node. JS. [9]

b) What is node. JS? Explain file handling in node. js. [8]

OR

Q4) a) Write a code using Express. JS to illustrate the concept of roots. [9]

b) What is replication? Enlist the advantages of replication in database system. [8]

P.T.O.

Q5) a) What is navigation? Write a code to navigate from one page to another page in jQuery mobile. [9]

b) What is jQuery mobile? Explain layout in jQuery Mobile design with its types. [9]

OR

Q6) a) Why CSS is important in designing mobile websites? Give an example of CSS class used for mobile website design. [9]

b) List any five widgets in jQuery mobile and explain any two in brief. [9]

Q7) a) What is EC2 service? Explain steps to deploy website on EC2. [9]

b) What is AWS cloud? What are the services provided by AWS? Explain any two in brief. [8]

OR

Q8) a) What is ELB? What are the ELB types? List advantages of ELB [9]

b) What is VPC? What are the components of VPC? [8]



Total No. of Questions : 8]

SEAT No. :

PC-1810

[Total No. of Pages : 2

[6353] - 129

TE-(Information Technology Engineering)
Web Application Development
(2019 Pattern) (Semester - II) (314453A)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2 , Q3 or Q4, Q5 or Q6 and Q7 or Q8
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) Explain event binding and property binding in angular with example [6]
- b) What is MVC? Explain MVC architecture in detail [6]
- c) Explain any 2 Hooks in React JS with example [6]

OR

- Q2)** a) What is TypeScript? Write typescript program which will find maximum and minimum number from array [6]
- b) Compare functional component with class component [6]
- c) What is Pipe? Explain with example [6]
- Q3)** a) What is NoSQL? Explain different features of MongoDB [5]
- b) Write a code using ExpressJS to illustrate the concept of routes [6]
- c) Explain the concept of Event Loop with suitable diagram [6]

OR

P.T.O.

- Q4)** a) What is API? Explain REST HTTP API's in detail [5]
b) Write code to create collection, display and search data in mongoDB using NodeJS [6]
c) Explain File system module NodeJS. Write program to perform append and delete operation on file [6]
- Q5)** a) List and Explain any 3 widgets in jQuery Mobile [6]
b) What is Mobile-First and Mobile Web? List different Mobile Devices [6]
c) What is navigation? Write code to navigate from one page to another in jQuery Mobile [6]

OR

- Q6)** a) Compare jQuery with jQuery Mobile [6]
b) Write jQuery Mobile code to hide button once it is clicked [6]
c) What is content? Write code to create content in JQuery Mobile page [6]
- Q7)** a) Explain working of Elastic Load Balancer in detail [6]
b) What are the different components of VPC? [5]
c) What is elastic beanstalk and how it works? [6]

OR

- Q8)** a) What are different AWS Storage types [6]
b) What is cloud computing? Enlist benefits of cloud computing [5]
c) What is elastic beanstalk and how it works? [6]



Total No. of Questions : 8]

SEAT No. :

PD4331

[Total No. of Pages : 2

[6403]-129

T.E. (IT)

WEB APPLICATION DEVELOPMENT

(2019 Pattern) (Semester - VI) (314453)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer four questions.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data, if necessary.*
- 4) *Neat diagram must be drawn wherever necessary.*

- Q1)** a) Explain how to create component and use it in angular application. [6]
b) Compare functional component with class component. [6]
c) Explain any 2 Hooks in ReactJS with example. [6]

OR

- Q2)** a) What is TypeScript? Write typescript program which will find maximum and minimum number from array. [6]
b) Explain any 2 types of Directive in Angular with example. [6]
c) Explain Angular - Architecture in detail. [6]

- Q3)** a) Compare NoSQL with SQL. [5]
b) Write a code using ExpressJS to illustrate the concept of routes. [6]
c) Explain the concept of Event Loop with suitable diagram. [6]

OR

- Q4)** a) What is API? Explain REST HTTP API's in detail. [5]
b) Write code to create collection, display and search data in MongoDB using NodeJS. [6]
c) Explain File system module NodeJS. Write program to perform append and delete operation on file. [6]

P.T.O.

- Q5)** a) List and Explain any 3 widgets in jQuery Mobile. [6]
b) Explain concept of 'role' in jQuery mobile? List and explain any 4 role. [6]
c) What is navigation? Write code to navigate from one page to another in jQuery Mobile. [6]

OR

- Q6)** a) Compare jQuery with jQuery Mobile. [6]
b) Write jQuery Mobile code to hide button once it is clicked. [6]
c) What is content? Write code to create content in Jquery Mobile page. [6]

- Q7)** a) Explain working of Elastic Load Balancer in detail. [6]
b) What is VPC? Explain advantages of VPC. [5]
c) What is PuTTY? How to connect EC2 instance with PuTTY? [6]

OR

- Q8)** a) What are different AWS Storage types? [6]
b) What is cloud computing? Enlist benefits of cloud computing. [5]
c) What is elastic beanstalk and how it works? [6]

